

Using integrated radiomic and pathomic-based models to predict progression-free survival in early-stage lung adenocarcinoma

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ABSTRACT

Early-stage non-small cell lung cancer (NSCLC) patients have a relatively high recurrence rate within the first five years of surgery, reflecting a need to predict post-surgical recurrence and offer personalized adjuvant therapies. Quantitative features extracted from radiology and pathology images can provide valuable information for the NSCLC recurrence prediction task, with radiomic features capturing global tumor phenotypes and pathomic features capturing local cellular and tumor microenvironment information. In this study, we propose to combine radiomic and pathomic features to predict progression-free survival within five years of curative resection in early-stage lung adenocarcinoma (LUAD), the most common subtype of NSCLC. Using 106 cases from the National Lung Screening Trial dataset, we extracted radiomic features from lung nodules on pre-surgery computed tomography (CT) scans guided by radiologist’s segmentation and pathomic features from hematoxylin and eosin (H&E)-stained whole slide images (WSIs) of the resected tissue. We leveraged both hand-crafted and deep features in each modality and used a Cox proportional hazards model.

Models were trained with 5-fold cross-validation with ten repetitions, and metrics such as the concordance index (C-index) was calculated by the mean performance on the test set. The fused model using combined radiomic and pathomic features has a C-index of 0.634. Our study shows that combining radiomic and pathomic features results in a more accurate progression-free survival prediction model as compared to only using radiomic features (C-index=0.612), pathomic features (C-index=0.584), or clinical features (C-index= 0.477).

Keywords: Non-small cell lung cancer, progression-free survival, radiomics, pathomics, data fusion

1. INTRODUCTION

NSCLC patients undergo surgery with curative-intent, but around 30-55% develop disease recurrence within the first five years of surgery.¹ Computational approaches that leverage existing data collected as part of standard of care to identify patients at the highest risk of post-operative recurrence are useful in informing decisions about adjuvant chemo and radiation therapies.

Images obtained from radiology and pathology hold significant potential in the prediction of cancer recurrence. For instance, radiology images inform macroscopic tumor phenotypes, such as shape and texture, while pathology images contain microscopic cellular phenotypes, such as tumor microenvironments. Previous work has explored radiomic and pathomic feature fusion for lung cancer recurrence prediction but has been limited to using hand-crafted features and binary classifiers that did not consider the effect of risk factors on the outcome over time.²

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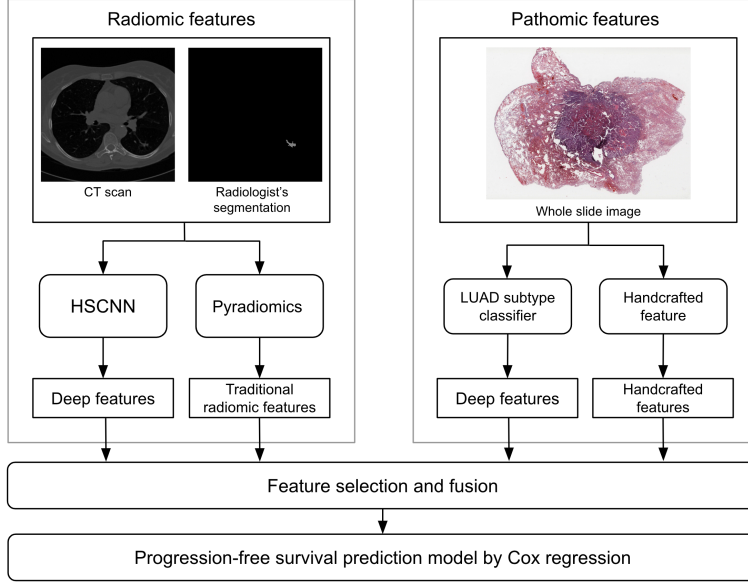


Figure 1. Flowchart illustrating the proposed method of using integrated radiomic- and pathomic-based models to predict progression-free survival in early-stage lung adenocarcinoma.

In this study, we propose utilizing information from radiomic and pathomic features to predict progression-free survival within five years of curative resection in early-stage LUAD, the most common subtype of NSCLC. Progression-free survival is defined as the time from surgical resection to disease recurrence or lung cancer death. Radiomic features were extracted from lung nodules on pre-surgery CT scans, and pathomic features were extracted from resection H&E-stained WSIs. Unlike prior works, we leverage both hand-crafted and deep features for radiomic and pathomic features and use a Cox proportional hazards model to consider the effect of risk factors on an outcome over time. Our study has shown that combining radiomic and pathomic features results in a more accurate progression-free survival prediction model as compared to when 1) using single modality radiomic or pathomic features, 2) using clinical features such as age and smoking status, and 3) using at least one clinical feature plus combined radiomic and pathomic features. The proposed progression-free survival prediction method is shown in Figure 1.

2. METHODS

2.1 Data Description

The CT scans and H&E-stained WSIs were obtained from the National Lung Screening Trial (NLST).³ We selected a cohort of 106 patients who had surgical resection as initial treatment for early-stage (stage I or II) LUAD. These patients have patient-level matched pre-surgery CT scans and resection WSIs, clinical information, and time to progression available. While a patient may have undergone up to three CT scans as part of the NLST, we analyzed the most recent scan available for each patient, as this scan was the closest to the surgical resection and the time when the tissue specimen was obtained.

2.2 Radiomic Feature Extraction

Two board-certified radiologists (D.A. and A.P.) and a fourth-year medical student (S.J.) jointly annotated the lung nodule contours on the CT scans at the 2-dimensional (D) slice level. The annotated contours were combined into a 3D volume, from which the radiomic features were extracted. A generative adversarial network-based CT denoising and normalization model was applied to the entire CT volume before radiomic feature extraction.⁴

Radiomic features consist of traditional features extracted using PyRadiomics⁵ and deep features extracted using a hierarchical semantic convolutional neural network (HSCNN),⁶ a nodule malignancy prediction model

that was developed and tested on 4252 nodules from the Lung Image Database Consortium (LIDC) dataset. For each nodule location, we provided the model with a 3D region of interest volume centered on that nodule and extracted 256 features from the last hidden layer.

Traditional PyRadiomics features were extracted from the intratumoral and peritumoral regions. The intratumoral region refers to the radiologist-outlined nodule region. The peritumoral region was obtained by dilating the nodule contour to a 20-mm peritumoral radius and subtracting the nodule contour from the dilated contour to generate a ring-shaped region. The dilation radius was based on the resection margin, which has a prognostic effect on lung cancer recurrence.⁷ Shape features were excluded from the peritumoral features. For each nodule, 107 intratumoral Pyradiomics features and 93 peritumoral Pyradiomics features were extracted.

In the case of multiple nodules in a patient, patient-level features were obtained by taking the mean of each feature across all nodules. The final patient-level features were a concatenation of 256 deep and 200 PyRadiomics features. This resulted in 456 radiomic features for each patient.

2.3 Pathomic Feature Extraction

Pathomic features include 512 deep features extracted from the last fully connected layer of a pretrained LUAD histologic subtype classifier (unpublished) and 27 hand-crafted features. The classifier is a tile-level ResNet18 classifier pretrained using self-supervised learning that was developed and evaluated on 610 patients from NLST (n = 146), The Cancer Genome Atlas (TCGA, n = 325), and The Clinical Proteomic Tumor Analysis Consortium (CPTAC, n = 139) cohorts. The average held-out test set F1 score for all subtypes across 5-fold cross-validation was around 0.9. The 512 features from the last hidden layer of ResNet18 were extracted. The hand-crafted features describe immune cell density and spatial colocalization with tumor cells within each tile.^{8,9} Tile-level pathomic features were aggregated into patient-level features by taking each feature’s mean, standard deviation, and range, resulting in 1617 pathomic features for each patient.

2.4 Survival Analysis

Features were individually selected via the Maximum Relevance Minimum Redundancy (mRMR) algorithm from each modality using the mRMRe R package.¹⁰ Within each modality, features were selected such that the chosen features are the most correlated with the target variable (progression status) and the least redundant. Feature selection was performed on the training set, and the same features were selected in the validation and testing sets. The selected radiomic and pathomic features were combined via vector concatenation.

We employed the Cox proportional hazards regression model with ridge regularization. The Cox model is a widely used statistical method in survival analysis that considers both censored and uncensored data. In our analysis, the selected features were treated as covariates, and their coefficients were estimated to determine their influence on the survival time and progression status of the individuals in our study population.

The optimal number of features from each modality and Cox model’s hyperparameters were determined based on the validation set C-index via event status-stratified 5-fold cross-validation with ten repetitions. Within each fold, there were 60% training, 20% validation, and 20% held-out testing patient cases. The final model performance comparison was done using the average C-index on the test set.

Table 1. Mean and standard deviation of C-index using repeated 5-fold cross-validation. (R, P) is the fused model with radiomic and pathomic features. (R, P, C) is the fused model with radiomic, pathomic, and clinical features. (R) is the radiomic-only model. (P) is the pathomic-only model. (C) is the clinical-only model.

Feature type	Training C-index	Validation C-index	Testing C-index
(R, P)	0.891 $\pm$ 0.026	0.631 $\pm$ 0.128	0.634 $\pm$ 0.130
(R, P, C)	0.892 $\pm$ 0.026	0.630 $\pm$ 0.128	0.627 $\pm$ 0.1236
(R)	0.785 $\pm$ 0.034	0.583 $\pm$ 0.142	0.612 $\pm$ 0.143
(P)	0.852 $\pm$ 0.038	0.634 $\pm$ 0.141	0.584 $\pm$ 0.149
(C)	0.627 $\pm$ 0.053	0.499 $\pm$ 0.146	0.477 $\pm$ 0.161

3. RESULTS

3.1 Combined Radiomic-Pathomic Model Achieves the Highest C-index

Table 1 summarizes the experiment results of the survival analysis. The combined radiomic-pathomic model achieved the best performance on the test set with a mean C-index of 0.634. The radiomic-only and pathomic-only models showed similar performance, with C-index values of 0.612 and 0.584, respectively. The clinical-only model had a C-index of 0.477. This is likely due to the less predictive capacity of these clinical features such as age, gender, pack-year, and pathologic staging. Notably, adding clinical features to the radiomic-pathomic model lowered its performance. Overall, compared to using a single modality, the radiomic-and-pathomic model obtained a higher mean and lower standard deviation in its test set C-index.

3.2 PyRadiomics Radiomic and Deep Pathomic Features Selected as Most Informative

The best radiomic-and-pathomic model uses 7 radiomic and 7 pathomic features. Throughout the repeated cross-validation process, on average 5.14 selected radiomic features were PyRadiomics features, and 1.86 were deep features. The most frequently selected radiomic features include PyRadiomics features from both the intratumoral and peritumoral regions. Among the 7 pathomic features, 6.14 were deep features, and 0.86 were hand-crafted features. The most frequently selected hand-crafted feature was the minimum value of the sum of pairwise distances between one immune cell and other cells.

4. CONCLUSION

Integrating radiomic and pathomic features results in a fused model outperforms other single-modality models and clinical feature-based models in predicting five-year progression-free survival. This multi-modal approach not only underscores the value of combining multimodal data for prognostication but also presents promising results toward informing treatment strategies in lung cancer care. In the future, other intermediate fusion techniques such as canonical correlation analysis can be explored, and datasets from other institutions will be added to further improve and validate the current model.

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